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The Inference Inflection: Where Real-Time AI Meets Real-World Opportunity

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While everyone was busy debating tariffs, AI inference quietly had its breakout moment.

While markets fixated on trade headlines and the daily churn of social media noise, a far more important macro trend was again quietly gaining momentum: AI. Long before tariffs took center stage, it was the release of DeepSeek R1 on Inauguration Day that truly set things in motion—triggering a wave of positioning unwinds driven by fears of capex spend and forcing investors onto the defensive by the time we reached Liberation Day.

While everyone sat on their hands waiting for the uncertainty to decline, capex fears were proven wrong and the tone of corporate Q1 earnings calls shifted decisively—from talk of model training and GPU arms races to the explosive growth of AI inference and capacity constraints. This is the real-time side of artificial intelligence, where trained models move from experimentation into action—powering decisions, automation, and scaled deployment.

Microsoft's Q3 FY25 earnings underscored the magnitude of this shift: Azure processed over 100 trillion tokens that quarter, including 50 trillion in March alone—a fivefold year-over-year surge. It marks a definitive pivot away from the research-heavy phase of AI and into the age of inference at scale.

And we're just getting started. With the upcoming rise of AI agents, AI-native phones and PCs, and embodied AI like self-driving platforms and humanoid robots, inference demand is poised to go parabolic. The implications for semiconductors, memory, networking, power, and edge compute are massive. If Q1 earnings told us anything, it's that scalable inference infrastructure has officially arrived.

Key Statements Highlighting the Inference Boom

The statements below, sourced from Q1 2025 earnings calls and summaries, emphasize the rapid ascent of inference as AI's next frontier and affirm that the time to act is now:

1. Microsoft (MSFT) – Q3 FY25 (January 2025)

"Inference usage has grown fivefold year-over-year, with over 100 trillion tokens processed, including a record 50 trillion in March alone, reflecting the shift toward real-time AI applications driving unprecedented demand for our Azure AI infrastructure."

– Satya Nadella, CEO

2. Amazon (AMZN) – Q1 2025 (April 2025)

"Most of the spend today in AI has been on training. But when you get to scale of the production applications for AI, it flips. The overwhelming majority of the spend is actually on inference—the predictions of the model."

– Andy Jassy, CEO

3. NVIDIA (NVDA) – Q4 FY25 (February 2025)

The CEO noted that inference has surpassed training as the primary AI workload in many applications, fueling demand for GPUs and networking solutions to support real-time processing in cloud and edge environments.

4. Meta Platforms (META) – Q1 2025 (April 2025)

The CEO highlighted that inference workloads now drive the majority of AI compute spending, as deployed models

power real-time features across products, requiring significant data center infrastructure upgrades.

5. Qualcomm (QCOM) – Q1 CY2025 (February 2025)

The CEO emphasized that AI inference at the edge is experiencing rapid growth, particularly in automotive and IoT, with chips enabling real-time processing for a multi-billion-dollar market.

6. Alphabet (GOOG) – Q1 2025 (April 2025)

“Our leadership in AI continues to drive extraordinary results in Google Cloud, where inference workloads are scaling rapidly, fueled by customer demand for real-time generative AI applications.”

– Sundar Pichai, CEO

7. AMD (AMD) – Q1 2025 (April 2025)

“What we’re seeing is that with these new reasoning models, inferencing is becoming more important. There’s also a shift toward more distributed inferencing, and I think that plays to our strengths.”

– Lisa Su, CEO

8. Equinix (EQIX) – Q1 2025 (May 2025)

The CEO stated that the surge in AI inference workloads is driving unprecedented demand for low-latency data center capacity, with hyperscalers expanding footprints to support real-time AI applications.

9. Micron Technology (MU) – Q3 FY25 (March 2025)

“Our HBM3E memory is seeing incredible demand for AI inference, where high bandwidth and low latency are critical, driving record data center revenue this quarter.”

– Sanjay Mehrotra, CEO

10. Intel (INTC) – Q1 2025 (April 2025)

The CEO noted that inference workloads are fueling demand for specialized AI chips and networking solutions, with their Gaudi accelerators and Ethernet offerings enabling real-time processing in data centers and edge environments.

11. Ciena Corporation (CIEN) – Q1 FY25 (March 2025)

The CEO noted that surging demand for AI connectivity is driving significant order growth from hyperscale customers, with high-bandwidth optical solutions enabling real-time AI applications like inference in data centers.

Inference has moved from the sidelines to the center of AI infrastructure. The following companies are broken down by strategic area—chips, memory, networking, data centers, cloud, and edge/embodyed AI—offering a roadmap for investors seeking exposure to the next major wave of AI-driven demand.

AI Compute & Inference Chips

These companies provide GPUs, accelerators, or edge inference chips:

1. NVIDIA (NVDA) – The market leader in both training and inference GPUs; inference now driving data center growth.
2. AMD (AMD) – Seeing momentum with MI300 and distributed inference models.
3. Intel (INTC) – Gaudi AI chips gaining traction; Ethernet and inference-optimized silicon part of roadmap.
4. Qualcomm (QCOM) – Dominant at the edge (phones, automotive, IoT); custom inference chips for low-power devices.
5. Lattice Semiconductor (LSCC) – FPGA-based low-power inference plays at the edge.

Memory & Storage

These names benefit from rising HBM, DRAM, and edge memory demand:

1. Micron Technology (MU) – Leader in HBM3E and low-latency DRAM used for inference workloads.
2. Western Digital (WDC) – Drives inference-related data center storage needs.

3. Seagate Technology (STX) – Supports growing edge/cloud storage requirements tied to inference.

Networking & Optics

Enabling inference via ultra-low latency, high-bandwidth connectivity:

1. Ciena Corporation (CIEN) – Optical backbone provider; strong AI data center build exposure.
2. Arista Networks (ANET) – Top player in high-speed networking for hyperscalers deploying inference.
3. Marvell Technology (MRVL) – Custom silicon and optical interconnects powering inference networks.
4. Broadcom (AVGO) – Dominant in networking chips, PCIe, and switching—core enabler of inference scale-out.

Data Centers & Infrastructure

Inference demand is straining and expanding data center infrastructure:

1. Equinix (EQIX) – Hyperscaler growth and latency-sensitive inference demand driving record colocation needs.
2. Digital Realty (DLR) – Benefiting from real-time AI workloads shifting inference into new facilities.
3. Vertiv (VRT) – Power and cooling infrastructure for inference-heavy data centers.

AI-Optimized Cloud Providers

Large-scale inference customers and beneficiaries of usage-based cloud economics:

1. Microsoft (MSFT) – Azure’s 100-trillion token quarter is the proof point.
2. Alphabet (GOOG) – Cloud inference usage expanding fast.
3. Amazon (AMZN) – AWS seeing a shift toward inference-driven demand.

Edge & Embodied AI Enablers

The next leg of inference: robotics, smart devices, FSD, and more:

1. Ambarella (AMBA) – Edge AI chips used in automotive and vision systems.
2. ON Semiconductor (ON) – Inference-capable image sensing and power semiconductors for AI-native systems.
3. NXP Semiconductors (NXPI) – Automotive AI and edge inference control.
4. Synaptics (SYNA) – AI-powered interfaces and inference at the edge (phones, consumer, IoT).

You’ll notice that many of the names on this list are semiconductor companies—and yes, they’re positioned to benefit the most. If you build an equal-weight basket of these stocks, the historical price chart closely resembles the SOX (Semiconductor Index). Each company addresses a different part of the inference buildout, but there’s no escaping the massive demand ahead for memory, networking, power, and edge compute as the AI race accelerates.

Many of these names are currently priced as if we’re heading into a cyclical downturn—partly due to tariff noise and capex fears stemming from DeepSeek. But ironically, DeepSeek has actually accelerated the shift toward inference over training, which is exactly why Satya Nadella’s recent comments were so important. He called DeepSeek a Jevon’s Paradox moment—and so far, he’s been right for increased demand.

As we move into the next phase—embodied AI—inference demand is only going to grow.

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